XENON INTERPRETABILITY

Prompt Confusion

Three-Family Conflict Geometry Checkpoint

April 16, 2026 — summary of the current prompt-local conflict benchmarks, shared geometry results, and the projection diagnostic that clarified the diversification story.

BOTTOM LINE

We now have three validated prompt-local conflict families: `trade_size`, `risk_preference`, and `diversification_preference`. All three are text-blind under lexical controls and linearly readable in the residual stream. The updated same-capture geometry story is that `trade_size` and `risk_preference` form the tightest pair, while `diversification_preference` is a real but somewhat more distant third member. The main caveat is no longer "is diversification real?" but "how much prompt-structure offset contaminates thresholded transfer?" This is valuable because it gives Concordance a real benchmark suite for multi-source policy reasoning inside an LLM, and it gives DX Terminal a concrete path toward detecting, explaining, and eventually steering cases where strategy guidance and user settings pull the model in different directions.

Background

The project started as an attempt to study how a trading-oriented LLM represents internal prompt conflicts between `STRATEGY` and `ACTIVE SETTINGS`. Several early designs were too confounded, semantically unstable, or dependent on temporal context the prompt did not actually contain. The main lesson from that process was methodological:

- strong conflict benchmarks need to be prompt-local
- conflict labels need clean aligned rows
- lexical controls need to be at chance
- and geometry claims should not rely on structurally broken families

The current checkpoint focuses on the three families that survived that filtering. It is primarily a datasets / analysis / results document, not a full retrospective on every discarded design.

Datasets

The current benchmark family set is:

Family	Rows	Aligned	Conflict	Decision axis

`trade_size` (Phase 09)	384	clean	very clean	output `size` field
`risk_preference` (Phase 10)	384	100% aligned exact	asymmetric	asset selection by risk posture
`diversification_preference` (Phase 12)	384	100% aligned exact	85.4% exact	asset selection by portfolio fit

Each family is a matched-pair benchmark with an aligned / conflict label. The families differ in what the disagreement is **about**:

- `trade_size` asks whether the model should buy `small` or `large`
- `risk_preference` asks which viable asset matches the allowed risk posture
- `diversification_preference` asks which viable asset best matches the allowed concentration vs broadening posture given the existing book

Two design decisions are worth highlighting.

First, `risk_preference` and `diversification_preference` both use asset selection rather than activity gating. That keeps the task prompt-local and lets us ask whether conflict representations generalize beyond a single output field.

Second, `diversification_preference` is explicitly portfolio-conditioned. `ALPHA` deepens an already-owned sleeve; `BETA` broadens into a more distinct one. This makes `PORTFOLIO` semantically operative rather than decorative.

Analysis

The analysis stack across these phases is now fairly consistent:

1. Behavioral smoke on aligned and conflict rows
2. Cheap and formal lexical gates on raw prompt text
3. Standard probe workflows with lexical holdouts
4. Marshall-style strict both-axes holdouts where available
5. Cross-family geometry and transfer
6. Same-capture projection / threshold diagnostics when transfer results look calibration-sensitive

All probe analyses here use:

- model: `Qwen3-30B-A3B`
- site: `resid_post`
- readout token: last prompt token
- layers sampled from `L0` to `L44`

The current geometry checkpoint relies on two complementary cross-family estimators:

- **TransferProbeSpec geometry**: logistic-regression probe directions transferred between families
- **same-capture mean-diff geometry**: family-specific conflict centroid differences computed inside one combined capture

The second estimator turned out to matter a lot for interpreting `diversification_preference`.

Results

1. Each family is independently real

trade_size remains the anchor benchmark. Its single-family probe results are near ceiling:

- XOR split: 0.9948 / 1.0000
- strategy holdout: 1.0000 / 1.0000
- settings holdout: 0.9948 / 1.0000

Behavioral compliance is also the strongest: ~97% follow-setting on conflict rows. This family is the baseline reference point for the project.

risk_preference is the second strong family. Its standard holdouts were:

- XOR split: 0.9635 / 0.9766
- strategy holdout: 0.9844 / 0.9937
- settings holdout: 0.9740 / 0.9839
- strict both-axes: 0.8854 / 0.9119

The main caveat is behavioral asymmetry on conflict rows: conservative constraints bind much more strongly than aggressive permissions.

diversification_preference is the third validated family. Its first-pass behavior is cleaner than risk:

- aligned rows: 1.0000 exact
- conflict rows: 0.8542 exact

And its probe results are strong:

- XOR split: 0.9896 / 0.9995
- strategy holdout: 1.0000 / 1.0000
- settings holdout: 0.9792 / 0.9957
- strict both-axes: 0.8333 / 0.8819

The strict result is softer than the easy splits, but still comfortably above chance with text baselines pinned at 0.500 / 0.500.

2. The original three-family geometry story was too pessimistic about diversification

The first three-family transfer workflow suggested:

- risk_preference and trade_size were the tightest pair
- diversification_preference was much farther away
- outgoing diversification transfer had high AUROC but often collapsed to 0.500 balanced accuracy

At L36, the original transfer-probe cosine picture was:

Pair	Cosine
risk vs trade_size	0.5341
diversification vs risk	0.2996
diversification vs trade_size	0.2846

Those numbers were real as probe-direction comparisons, but they mixed shared conflict structure with family-specific classification features. That mattered most for diversification.

3. The projection diagnostic clarified the geometry

The follow-up projection diagnostic recomputed family-specific conflict directions inside the **same combined capture** and then evaluated all families against those directions.

At `L36`, the updated mean-difference cosine matrix is:

Pair	Cosine
`risk vs trade_size`	`0.6449`
`diversification vs risk`	`0.4684`
`diversification vs trade_size`	`0.4883`

This changes the diversification story materially.

The right picture now is not:

- risk and size are related
- diversification is barely connected

It is:

- risk and size are the tightest pair
- diversification is a real third member
- but it sits on a stronger family-specific baseline offset

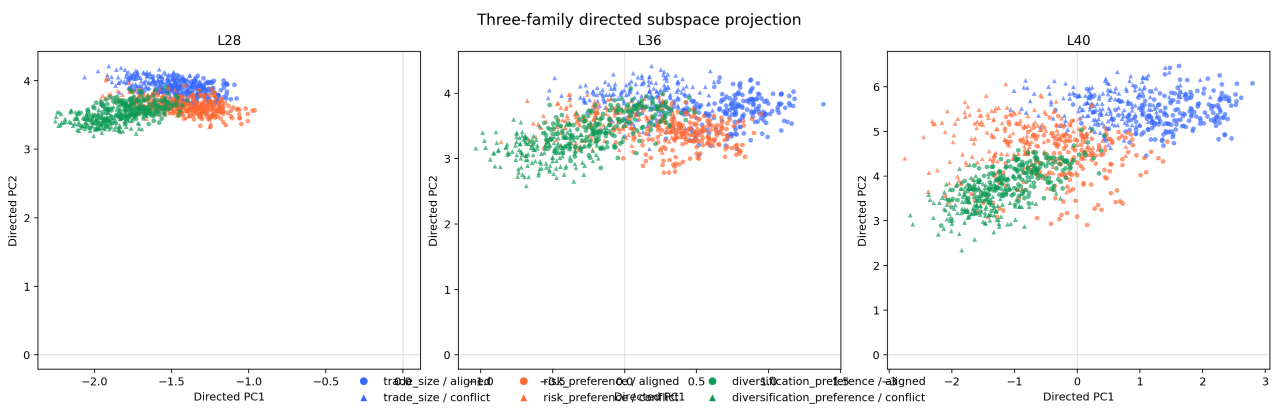

Figure 1: Directed 2D projection of all 1152 rows onto the conflict-relevant subspace derived from the three same-capture mean-diff directions. `L36` gives the clearest overall view: risk and trade size are the tightest pair, while diversification is visibly related but shifted.

4. The big diversification problem is threshold transfer, not absence of signal

The projection diagnostic also answered why diversification looked so odd under cross-family transfer.

On the diversification direction at `L36`, the within-family score bands are:

- aligned mean: `-2.1717`
- conflict mean: `-1.6130`
- threshold: `-1.8923`

Applying that same diversification direction to other families gives decent ranking but broken threshold transfer:

- diversification -> `trade_size`:
  - AUROC `0.8471`

  - balanced accuracy `0.5000`
  - FNR `1.0000`
- diversification -> `risk_preference`:
  - AUROC `0.8165`
  - balanced accuracy `0.5599`
  - FNR `0.8802`

The reverse direction shows the complementary effect:

- `risk_preference` -> diversification:
  - AUROC `0.8185`
  - balanced accuracy `0.5000`
  - FPR `1.0000`
- `trade_size` -> diversification:
  - AUROC `0.8302`
  - balanced accuracy `0.5859`
  - FPR `0.8281`

That is the signature of an additive baseline shift:

- scores preserve ordering well enough for AUROC
- but the source-family threshold is badly miscalibrated on the target family

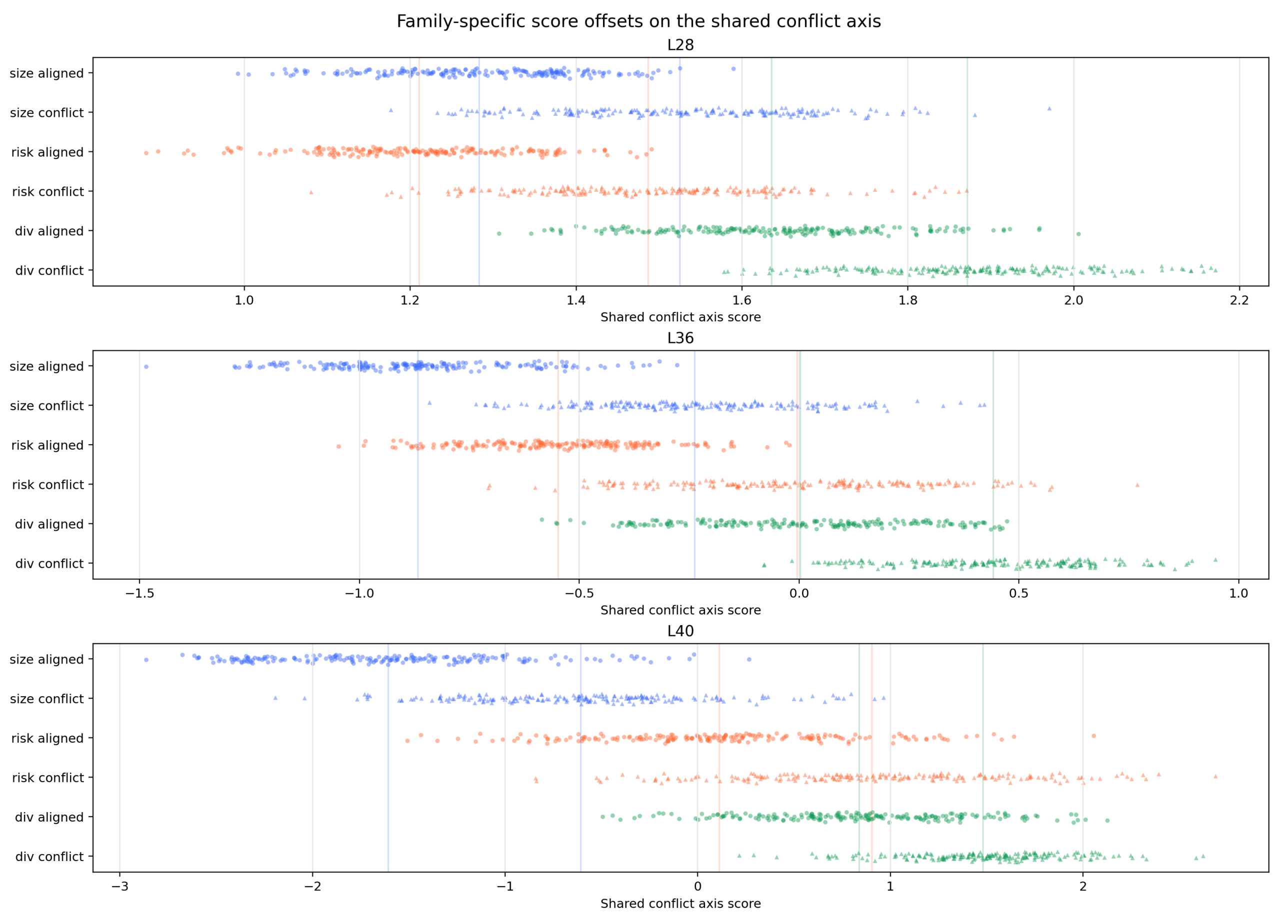

Figure 2: The most informative current figure. All three families separate aligned from conflict rows on the shared axis, but their overall baselines are shifted relative to one another. This explains why cross-family AUROC can stay high while thresholded balanced accuracy collapses.

The likely source is prompt structure. `diversification_preference` uniquely includes a semantically active `PORTFOLIO` block, and because we are reading the last prompt token, that extra section appears to shift the prompt-EOS residual in a way that projects onto the conflict direction.

Implications

The project-level claim is now stronger and cleaner than it was a few days ago.

> **CURRENT CLAIM**
>
> Qwen3-30B-A3B builds readable internal representations of strategy/settings disagreement across multiple prompt-local execution dimensions. `trade_size`, `risk_preference`, and `diversification_preference` are all independently decodable, text-blind under lexical controls, and linked by shared conflict geometry (cross-family transfer AUROC up to `0.98`). `risk_preference` and `trade_size` form the tightest pair; `diversification_preference` is a real third member with a larger family-specific baseline offset. The main remaining external-validity question is transfer to real DX Terminal prompts and other models.

Why the geometry matters

A probe trained entirely on size conflict can detect risk conflict at `0.98` AUROC — near-perfect ranking transfer. The model is not building three unrelated detectors. It is building related features that share substantial structure.

That is what the geometry work adds on top of the individual probe results. Probe accuracy says conflict is readable on each axis. The cross-family transfer says the underlying signal is structured enough to generalize across execution dimensions.

That matters because it changes how we can apply the result:

- for Concordance, it gives us a reusable benchmark and subspace for testing monitoring, transfer, and causal tooling
- for DX Terminal, it gives us a path to detect when strategy and settings are in tension, identify **which axis** is in tension, and distinguish missing conflict detection from strange downstream resolution

Three concrete implications follow, and they matter at two levels: the Concordance research stack and the DX Terminal product.

First, we should stop treating thresholded cross-family balanced accuracy as the main geometry metric. For these cross-family comparisons, the more honest signals are:

- same-capture mean-diff cosine
- cross-family AUROC
- explicit score-offset / threshold-shift diagnostics

Second, no additional synthetic family is needed right now. The geometry claim already has enough structure:

- one tight pair (`risk_preference`, `trade_size`)
- one moderately related third member (`diversification_preference`)
- one well-understood calibration caveat (`PORTFOLIO` offset)

Third, the project is now better positioned for **qualitative upgrades** rather than simple family expansion. The natural next directions are:

- real-data transfer, probably beginning with `trade_size`
- activation patching once the infrastructure is ready
- a shareable combined report that anchors the geometry story with the new projection figures

Why this is valuable to Concordance

For Concordance, this upgrades prompt-conflict analysis from an anecdotal debugging exercise into a reusable interpretability benchmark.

What we have now is not just a single interesting probe, but a benchmark family with:

- prompt-local labels
- text-blind lexical controls
- matched-pair behavioral grounding
- shared cross-family geometry
- and a clear diagnostic for separating shared signal from prompt-structure offset

That matters to Concordance in three ways.

First, it is a credible internal benchmark for evaluating whether model representations preserve distinctions that matter in instruction-composed systems.

Second, it is a testbed for new mechanistic tooling. The next causal or transfer method we build can be checked against a benchmark where the geometry is already partially mapped, instead of being tried first on messy live data.

Third, it is a concrete product-facing demo of what Concordance is for: turning invisible prompt interactions into measurable internal structure, rather than only scoring outputs after the fact.

Why this is valuable to DX Terminal

For DX Terminal, the value is operational.

DX Terminal already asks the model to combine multiple policy sources:

- strategy preferences
- active settings
- market information
- and sometimes portfolio state

The current result says the model does not merely emit a final answer; it builds readable internal representations of **where those policy sources disagree**. That matters because many product failures in a system like DX Terminal are not raw capability failures. They are policy-composition failures:

- the strategy says one thing
- the settings imply another
- the user sees a surprising execution choice
- and it is hard to tell whether the model missed the conflict, noticed it but resolved it strangely, or got nudged by prompt structure

This work gives us the technical foundation to build toward a DX Terminal workflow where we can:

- flag prompts whose internal conflict score is unusually high
- explain which execution axis is in tension (`size`, `risk`, or `diversification`)

- distinguish “the model did not see the disagreement” from “the model saw it but resolved it in a product-surprising way”
- and eventually intervene or rewrite prompts before a bad execution reaches the user

One product-facing lesson is already visible in the benchmark family: settings phrased as hard constraints bind more reliably than settings phrased as permissions. The exact behavioral asymmetry differs by family, but the broader pattern is stable enough to matter for DX Terminal prompt design. If we want settings to actually control execution, restrictive wording is much more trustworthy than permissive wording.

More broadly, the product payoff is better observability into policy alignment inside the prompt, not just better hindsight after an odd trade recommendation has already appeared.

Recommendations

Near-term:

- use the new same-capture visuals as the anchor figures for team discussion
- treat `trade_size`, `risk_preference`, and `diversification_preference` as the settled benchmark family set
- do not spend more cycles inventing additional synthetic settings families

Next experiments:

- apply the cleanest family directions to real DX Terminal prompts, starting with `trade_size`
- return to activation patching when the new infra is stable

This checkpoint gives us a coherent multi-family benchmark story with one clean anchor (`trade_size`), one strong but asymmetric family (`risk_preference`), and one portfolio-conditioned family (`diversification_preference`) whose geometry now looks interpretable rather than anomalous.